

CSCE 679 DATA VISUALIZATION · MODULE 09 · LECTURE NOTES

Uncertainty

The thing almost always omitted and almost always decisive.

MODULE OUTCOMES

- Explain the kinds of uncertainty and their sources.
- Explain the representations and what each communicates.
- Explain why error bars are misread.
- Explain the frequency framings that work better.
- Show uncertainty in a figure and test the reading.

1 Why uncertainty is omitted

UNCERTAINTY IS OMITTED BECAUSE IT IS HARDER TO DRAW AND BECAUSE IT WEAKENS THE FIGURE

A point estimate is one mark and a distribution is several — so showing the uncertainty costs channels, space, and visual clarity, and a cluttered figure is a real cost rather than an excuse (Module 02 §3's emphasis budget).

And it weakens the apparent finding, which is the uncomfortable half of the explanation: a trend drawn with its confidence band is visibly less convincing than the same trend drawn alone, and the author knows it.

Which means the omission is frequently an honest decision with a dishonest effect — the author was simplifying for legibility, and the reader concluded something the data does not support, and nobody intended the outcome.

So the position this module takes is: a point estimate drawn without its uncertainty invites a conclusion the data does not support, and that makes showing it a requirement rather than a refinement — which is why Project 2 fails a figure that omits it.

Kind	Where it comes from	Shown as
Sampling uncertainty	You measured a sample rather than the whole population.	A confidence or credible interval.
Measurement uncertainty	The instrument or the method is imprecise.	An error bar, or a band around the series.
Model uncertainty	The fit could have been different given the same data.	A band, or several fitted draws overlaid (§2).

Kind	Where it comes from	Shown as
Missing data	Some values are simply absent.	Explicitly — a gap in the line, not an interpolation across it. See the note.
Forecast uncertainty	The future has not happened yet.	A widening fan, with the width honestly derived rather than chosen for appearance.

Missing data shown as an interpolated line is the quietest of these failures — the figure asserts values that were never measured, and nothing in it says so. A series with a six-month gap drawn as a continuous line has invented six months of data, and the reader has no way to know. **The interpolated-gap failure is specific and worth naming**, because the fix is a single argument in every plotting library and is simply not reached for.

2 Error bars are misread

READERS DO NOT KNOW WHAT AN ERROR BAR MEANS, AND THE CONVENTION DOES NOT SAY

An error bar may show one standard deviation, one standard error, a 95% confidence interval, the interquartile range, or the full range — and these differ by factors of several on exactly the same data, so the bar's length carries no interpretable meaning on its own.

So a figure with unlabelled error bars is not interpretable, and a large fraction of published figures do not say which quantity they used — which is a one-clause omission that makes the figure's central visual element meaningless.

And readers misread even the labelled ones: they treat non-overlapping bars as indicating a significant difference and overlapping ones as indicating none, which is wrong in both directions (the relationship depends on which quantity is plotted and on the test) and is a well-measured finding.

Plus within-the-bar bias — readers judge a point just inside the bar as substantially more likely than a point just outside it, as though the interval had a hard boundary, which it does not: the density is continuous across the end of the bar.

- A gradient or a violin, which shows that the density falls off smoothly rather than implying a boundary — and so directly addresses the within-the-bar bias.
- Several draws from the posterior or from a bootstrap — spaghetti plots — which show what the plausible alternative fits actually look like, and communicate the shape of the uncertainty rather than its width.
- Hypothetical outcome plots: animate through individual plausible datasets, one at a time, which readers interpret measurably better than static intervals — one of the few places animation wins (Module 06 §3's exception).
- Quantile dotplots — twenty dots placed at the distribution's quantiles, so that the reader counts rather than estimates, which is §3's frequency framing made visual.

- **And explicit gaps for missing data, which is the cheapest of these and the most often skipped. Quantile dotplots let the reader count rather than estimate — and counting is a far more reliable operation than reading a density,** which is why they outperform both bars and bands in testing.

3 Frequency framing

PEOPLE READ FREQUENCIES BETTER THAN PROBABILITIES, WHICH IS A DESIGN OPPORTUNITY

'3 out of 20' is understood more accurately than '15%', consistently, across populations and across levels of numeracy — and a figure showing twenty dots with three of them highlighted is understood better still.

Which is exactly why quantile dotplots work: twenty dots positioned at the quantiles of the predictive distribution, so that the reader counts how many fall beyond a threshold rather than integrating a density function by eye, which nobody does well.

And it transfers to any threshold question: 'will this exceed the limit' becomes 'how many of these twenty are above the line' — which is a counting task rather than an estimation task, and counting is accurate.

So where the reader has an actual decision to make, frame the uncertainty as a frequency — which is CSCE 671 Module 01's task-first framing applied to uncertainty, and is the single most useful finding in this module for anybody building something people will act on.

4 Practice

- Say which quantity the interval represents, always — standard error, standard deviation, 95% interval, range — which costs one caption clause and is omitted constantly (§2).
- Prefer a representation without a hard boundary (§2's list), since the boundary is precisely the thing readers over-read — a gradient or a dotplot in place of a capped bar.
- Show the sample size, per group — which both a box plot and an error bar hide completely, and which determines how much the interval means: an interval from $n = 4$ and one from $n = 400$ look identical and are not.
- Never interpolate across missing data silently, which asserts measurements that do not exist (§1's note) — break the line, and say in the caption why.
- And for small denominators, show the uncertainty rather than the point estimate (Module 08 §3's small-denominator problem) — or consider a shrinkage estimate, stated as such. Showing the sample size per group is the cheapest uncertainty information available, and both the box plot and the error bar conceal it.

AND THE HONEST SUMMARY

Showing uncertainty makes a figure less persuasive and more honest, and those two pull in opposite directions whenever anybody wants the figure to persuade — which is most of the time, and is the real explanation for §1.

Which is why it is omitted, and why omitting it is this subject's commonest *honest* failure — as distinct from Module 10's deliberate ones, which are rarer and easier to condemn.

And the representations have genuinely improved: gradients, posterior draws, quantile dotplots, and hypothetical outcome plots all read measurably better than error bars, which were a convention inherited from print rather than a design.

So there is no longer a good excuse — the techniques exist, they are implemented in every major library, and they read better — which leaves only the persuasiveness argument, and that one is worth naming out loud when it is what is actually operating.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Wilke, the uncertainty chapters (free)	The whole module , free — and the best available practical treatment, with every representation drawn.
Hullman et al. — Hypothetical Outcome Plots (free)	§2's animated representation, measured — and it reads better than static intervals, which is the finding.
Kay et al. — When (ish) is My Bus? (free)	§3's quantile dotplots in the original — the frequency framing applied to a real decision.
Belia et al. — Researchers misunderstand confidence intervals (free)	§2's misreading, measured on researchers — which is the relevant population and the sobering result.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Take a figure you made** and add the uncertainty, then compare how persuasive each looks.
- 2 **Classify the uncertainty in one dataset** into the five kinds.
- 3 **Find a figure with unlabelled error bars** and compute what three different interpretations would mean.
- 4 **Ask five people whether two overlapping error bars indicate a significant difference.**

- 5 **Draw the same uncertainty as a bar, a gradient, and a quantile dotplot**, and test the readings.
- 6 **Build a spaghetti plot** from bootstrap draws.
- 7 **Pose a threshold question** and compare a percentage against a twenty-dot framing.
- 8 **Find a series with missing data drawn as continuous**, and fix it.
- 9 **Add the per-group n** to a box plot.
- 10 **Find a small-denominator extreme** on a map and show its interval.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Give the two reasons uncertainty is omitted, and why the omission is usually honest.
- 2 State the position this module takes.
- 3 Name five kinds of uncertainty and how each is shown.
- 4 Which failure is quietest, and why?
- 5 Why is an unlabelled error bar uninterpretable?
- 6 Give two ways readers misread error bars.
- 7 Name five better representations.
- 8 Why do quantile dotplots work?
- 9 State the frequency finding and how to use it.
- 10 Give five practice rules, and the cheapest one.